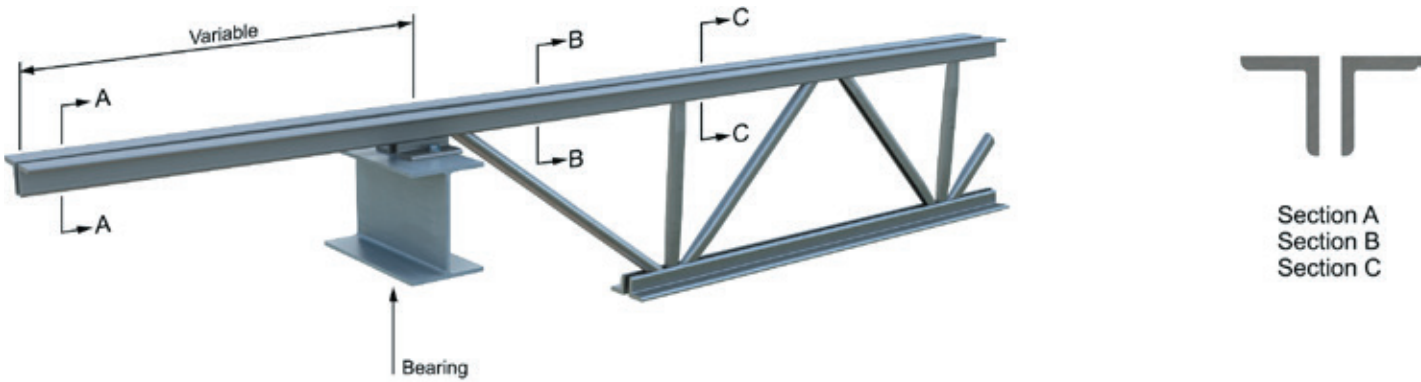
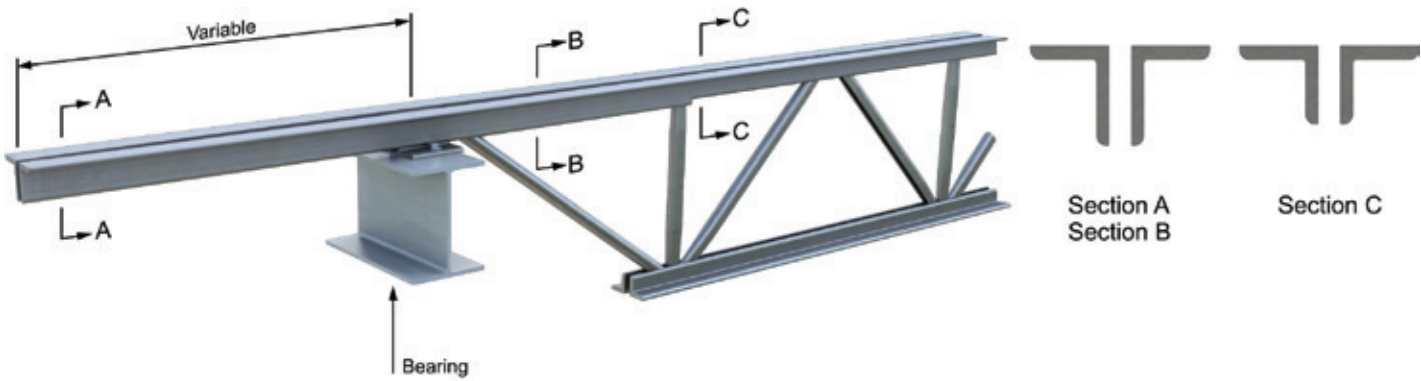


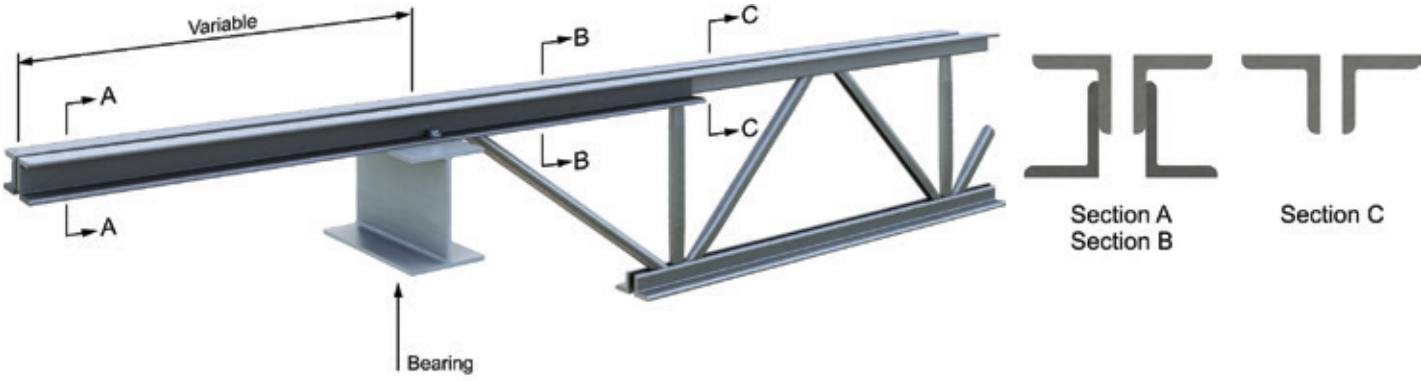
Standard details



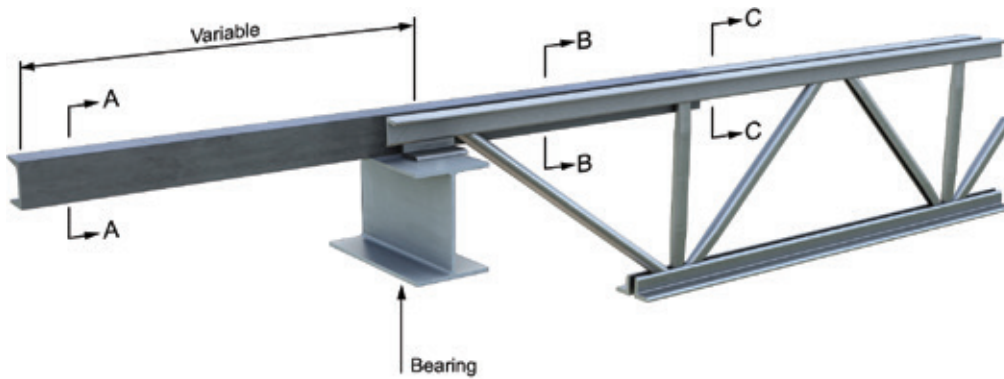
Top chord extension, section without reinforcement



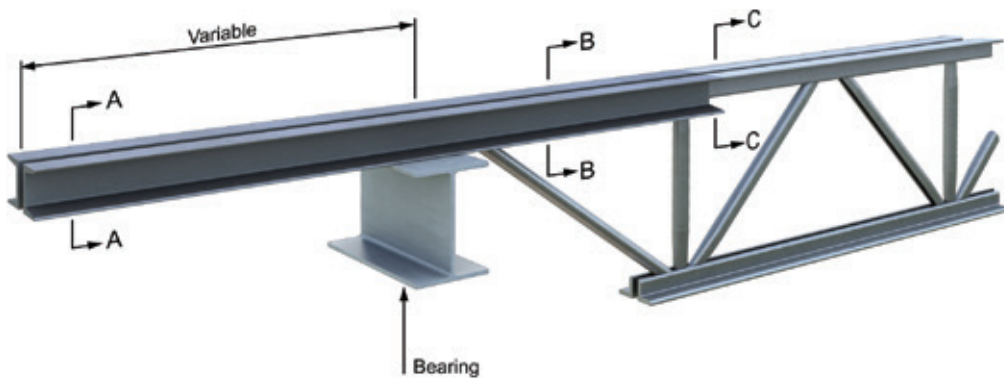
Top chord extension, section reinforced with 2 angles



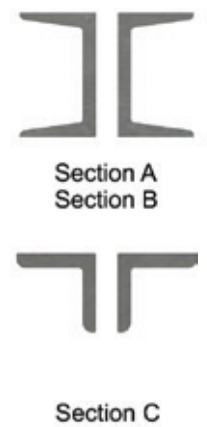
Top chord extension, section reinforced with 4 angles



Top chord extension, section reinforced with 1 channel



Top chord extension, section reinforced with 2 channels



Standard details

The tables below serve as a guide to determine a suitable shoe depth based on uniform loading and a maximum extension length. The extensions are based on the maximum capacity of a two-channel section without any slope. This is an economical section for this kind of condition.

The maximum top chord extension is determined by the bending and shear resistance of the section, or by the deflection of the extension, which is limited to $L/120$ with a fixed end at joist bearing point. In fact, the joist and its extension are analyzed simultaneously in a matrix calculation.

MAXIMUM TOP CHORD EXTENSION (mm)

Effective shoe depth (mm)	Factored load (kN/m)												
	Service load (kN/m)												
	4.5	6.0	7.5	9.0	10.5	12.0	13.5	15.0	16.5	18.0	19.5	21.0	22.5
	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0	11.0	12.0	13.0	14.0	15.0
100	1,920	1,750	1,620	1,520	1,450	1,380	1,330	1,290	1,240	1,200	1,150	1,130	1,100
125	2,390	2,170	2,010	1,900	1,800	1,700	1,650	1,550	1,500	1,450	1,400	1,350	1,300
150	2,750	2,500	2,350	2,200	2,050	1,950	1,900	1,800	1,750	1,650	1,600	1,550	1,550
175	3,050	2,800	2,600	2,450	2,300	2,200	2,150	2,050	2,000	1,900	1,850	1,800	1,750
200	3,300	3,000	2,800	2,650	2,550	2,450	2,350	2,250	2,200	2,100	2,050	2,000	1,950

MAXIMUM TOP CHORD EXTENSION (ft.)

Effective shoe depth (in.)	Factored load (lb./ft.)												
	Service load (lb./ft.)												
	300	405	510	615	720	825	930	1035	1140	1245	1350	1455	1560
	200	270	340	410	480	550	620	690	760	830	900	970	1040
4	6' - 4"	5' - 9"	5' - 4"	5' - 0"	4' - 9"	4' - 6"	4' - 4"	4' - 3"	4' - 1"	3' - 11"	3' - 9"	3' - 8"	3' - 7"
5	7' - 10"	7' - 1"	6' - 7"	6' - 3"	5' - 11"	5' - 7"	5' - 5"	5' - 1"	4' - 11"	4' - 9"	4' - 7"	4' - 5"	4' - 3"
6	9' - 0"	8' - 2"	7' - 8"	7' - 3"	6' - 9"	6' - 5"	6' - 3"	5' - 11"	5' - 9"	5' - 5"	5' - 3"	5' - 1"	5' - 1"
7	10' - 0"	9' - 2"	8' - 6"	8' - 0"	7' - 7"	7' - 3"	7' - 1"	6' - 9"	6' - 7"	6' - 3"	6' - 1"	5' - 11"	5' - 9"
8	10' - 10"	9' - 10"	9' - 2"	8' - 8"	8' - 4"	8' - 0"	7' - 8"	7' - 4"	7' - 3"	6' - 11"	6' - 9"	6' - 7"	6' - 5"

The designer must provide a shoe depth sufficiently high when the top chord is not horizontal to prevent conflict with top flange of the beam. In this case, the clear depth for the top chord is less than the shoe depth.

